MOBILE COMPUTING

ASSIGNMENT 1 – CO1

Design, Analyse and Optimise a Hexagonal Cellular Network

for Spectrum Allocation, Interference Management and Scalable Coverage Planning

Student Details	Information
Name	A. Manoj Reddy
Register Number	192311171
Course	Mobile Computing (ITA03)
Assignment	Assignment 1 – CO1
Total Marks	50

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Design Basis and Assumptions

The assignment specifies a 12 km² university township, 336 duplex voice channels after control-channel overhead, busy-hour traffic of 0.025 Erlang per subscriber, a target blocking probability no greater than 2%, subscriber scenarios of 1,000, 10,000, 20,000 and 100,000+, path-loss exponent n between 2.5 and 4, and a minimum SIR threshold to be selected between 12 dB and 18 dB.

Planning assumptions used for a worked design: N = 7 (valid hexagonal cluster with i = 2, j = 1); nominal cell radius R = 0.5 km; representative path-loss exponent n = 3.5 for a mixed campus/residential environment; minimum design SIR = 18 dB as a conservative target; six first-tier co-channel interferers for an omni-directional idealised layout. Real deployment requires a drive-test/propagation survey and detailed traffic map. The numerical values below are analytical planning estimates, not measured field results.

Important modelling note: 336 is the total system channel inventory, not 336 channels in every cell. The average allocation per cell in an N = 7 reuse cluster is 336/7 = 48 channels. Actual integer channel allocation and Erlang-B blocking must be checked per cell or per sector.

Executive Recommendation

Use a planned N = 7 reuse pattern with three-sector sites in dense zones, macro coverage for broad-area continuity, and demand-triggered microcells/small cells at high-load locations. Apply dynamic channel assignment or controlled channel borrowing only where interference coordination permits. Do not permanently redesign the whole township for a four-hour event; deploy temporary stadium capacity (temporary small cells or a cell-on-wheels) and sectorised stadium coverage, with pre-event optimisation and post-event removal.

A single N = 7 macro layer is a sound starting point for coverage and interference control, but it cannot serve the 100,000-subscriber forecast or the event surge with the fixed 336-channel inventory. The design must obtain more spatial reuse through smaller cells, sectorisation, and load distribution; no radio-planning method can create additional spectrum.

TASK 1 – Design Alternative Frequency-Planning Strategies (5 Marks)

Three distinct strategies are compared. Each uses the same finite spectrum but changes how channels are spatially reused.

Strategy A: Fixed Cluster Frequency Reuse (iterative)

Divide the hexagonal grid into clusters of N cells. Each cluster receives all 336 channels partitioned into N disjoint groups; a cell uses one group, and the pattern repeats. For hexagonal geometry, valid cluster sizes are $N = i^2 + ij + j^2$, where i and j are non-negative integers. This is spatial reuse. A practical first case is N = 7 (i = 2, j = 1), giving approximately 48 channels per cell.

Access concept: in a legacy FDMA/TDMA system, each assigned channel or time-slot resource is reserved for the cell; control channels are separated from traffic channels in the stated 336-channel net inventory. In an OFDMA system, the same principle is implemented with frequency-time resource blocks and coordinated reuse.

```
for each valid (i, j):\n    N = i*i + i*j + j*j\n    compute reuse distance D = R*sqrt(3*N)\n    estimate SIR and channels_per_cell =
```

Why investigate: it is simple, predictable and inexpensive to plan. It is a baseline against which adaptive methods can be compared. Preprocessing: subscriber-density/traffic map, hex-grid/site layout, terrain and building data, and propagation survey. The operator bears radio-planning and site-survey costs; the university may provide access and site permissions.

Strategy B: Sectorised Frequency Reuse (spatial filtering)

Use directional antennas at each site to divide a cell into three 120° sectors (or six 60° sectors). The same channel group can be reused at a shorter distance in appropriately oriented sectors because antenna patterns suppress many interferers outside the serving sector. For a first-order 120° model, the dominant first-tier co-channel interferer count may fall from six to about two; actual count and suppression depend on antenna patterns, tilt, azimuth and site geometry.

```
for each site:\n    divide coverage into 3 (or 6) directional sectors\n    assign channel subsets to sectors with azimuth coordinatio
```

Design principle: spatial filtering. It improves SIR without increasing licensed spectrum, but adds antennas, radio chains, optimisation and handoff boundaries. Preprocessing: sector traffic map, antenna pattern data, azimuth/tilt survey and propagation measurements. The operator bears equipment and optimisation costs.

Strategy C: Hierarchical Macro–Micro Layer with Dynamic Assignment / Borrowing (recursive)

Keep a macro layer for continuous coverage and mobility. Recursively split overloaded areas into microcells/picocells; allocate channels dynamically from a coordinated pool, or borrow idle channels from neighbouring cells subject to an interference check. The design principle combines spatial reuse and trunking efficiency. It is recursive because the same split-and-replan operation is repeated only in overloaded regions.

```
plan macro grid and measure busy-hour load\nfor each overloaded cell:\n    add microcell(s) and reduce local cell radius\n    partiti
```

Access concept: dynamic assignment is compatible with channelised FDMA/TDMA; in LTE/5G-like OFDMA, scheduling and resource-block coordination provide an analogous mechanism. Control signalling remains protected; traffic resources are assigned according to load. Preprocessing: time-varying traffic heat map, candidate-site survey, backhaul/power availability, mobility and handoff measurements. The operator bears survey, site, backhaul and operations costs.

Strategy comparison

Strategy	Sharing method	Main principle	Strength	Limitation
A. Fixed N reuse	Static disjoint channel groups repeated every N cells	Spatial reuse	Simple, predictable baseline	Fixed allocation can strand idle channels
B. Sectorised reuse	Directional sector channel plans	Spatial filtering	Higher SIR / potentially tighter reuse	Antenna cost and sector handoffs
C. Macro–micro + dynamic	Adaptive allocation and selective cell splitting	Trunking + spatial reuse	Targets hotspots and changing demand	Highest planning/coordination complexity

TASK 2 – Identify the Governing Design Parameter (2 Marks)

Strategy	Dominant parameter	Why it governs design	Growth response
Fixed reuse	Cluster size N (or Q = D/R)	N sets channels per cell and co-channel separation	Keep N only as large as SIR requires; add cells to raise spatial reuse

Strategy	Dominant parameter	Why it governs design	Growth response
Sectorised reuse	Effective co-channel interferer count and antenna discrimination	Directional gain changes interference sum and outage	Increase sector coordination / improve antenna patterns as load grows
Macro–micro + dynamic	Cell radius R and offered load per cell; available pool occupancy	R controls area/sites and splitting; pool occupancy controls blocking	Reduce R selectively and dynamically rebalance channels

For fixed spectrum, increasing N reduces channels per cell approximately as 1/N but increases reuse distance and SIR. Thus N is the central trade-off. In sectoring, the effective interference geometry is dominant. In the hierarchical strategy, the hotspot radius and the fraction of overloaded cells determine how many new sites and handoffs are required. Subscriber growth cannot be handled by increasing channels; it requires more spatial reuse and better traffic distribution.

TASK 3 – Derive Capacity and Interference Functions Mathematically (6 Marks)

3.1 Hexagonal geometry and cluster size

A hexagonal cluster is indexed by integers i and j. Moving i cells along one lattice axis and j cells along an axis 60° away gives the cluster relation $N = i^2 + ij + j^2$. For i = 2, j = 1: N = 4 + 2 + 1 = 7.

The co-channel reuse distance is $D = R\sqrt{(3N)}$. Therefore, with R = 0.5 km and N = 7: $D = 0.5\sqrt{21} = 2.291$ km; $Q = D/R = \sqrt{21} = 4.583$.

3.2 Cell area and number of cells

For a regular hexagon with distance R from centre to vertex, side length is R and area is $A_{cell} = (3\sqrt{3}/2)R^2$. With R = 0.5 km, $A_{cell} = 1.299 \times 0.25 = 0.650$ km². The ideal number of cells is 12/0.650 = 18.47, so at least 19 cells are required geometrically. Allowing for boundary overlap, roads, terrain and capacity, a practical plan may need more sites.

3.3 Channels and Erlang-B traffic capacity

The 336 net channels are divided across N = 7 cells: $S_{cell} = S/N = 336/7 = 48$ channels per cell (before any further sector partition). For m channels and offered traffic A Erlangs, Erlang-B blocking is $B(A,m) = [A^m/m!]/\Sigma(k=0\ldots m)\ A^k/k!$. It can be evaluated stably by recurrence $B_0 = 1$ and $B_m = A\,B_{(m-1)}/(m + A\,B_{(m-1)})$. The admissible cell load A_{max} is the largest A for which $B(A,48) \le 0.02$. The exact value should be obtained with an Erlang-B calculator or the included code; it is roughly in the mid-30 Erlangs (the code calculates it exactly).

Township offered traffic is A_{total} = subscribers × 0.025 Erlang: 25, 250, 500 and 2,500 Erlangs for 1k, 10k, 20k and 100k subscribers. With 19 cells, average offered load per cell is approximately 1.32, 13.16, 26.32 and 131.58 Erlangs. The 100k case exceeds the approximate 2%-blocking capacity of a 48-channel cell, even before uneven traffic distribution.

Subscribers	Total offered traffic (Erlangs)	Average per 19 cells (Erlangs)	Interpretation
1,000	25.0	1.32	Comfortable average load
10,000	250.0	13.16	Manageable average; hotspots still matter
20,000	500.0	26.32	Monitor busy cells
100,000	2,500.0	131.58	Average cell likely overloaded; densification required

3.4 SIR derivation

Assume equal transmit powers, path loss proportional to d^{-n}, a mobile at the cell edge, and six first-tier co-channel interferers each approximately D away. Desired signal is proportional to R^{-n}; each interferer is proportional to D^{-n}. Thus $SIR \approx (D/R)^n / 6 = (\sqrt{(3N)})^n / 6$. In dB, $SIR_{dB} = 10 \log_{10}[(3N)^{(n/2)}/6]$. For N = 7 and n = 4, SIR ≈ 18.7 dB, matching the assignment sanity-check. For the mixed-environment n = 3.5, the same approximation gives about 16.1 dB, below the selected 18 dB conservative target. Therefore the unsectored N=7 plan alone is not sufficient under this n assumption; sectoring/antenna discrimination or a larger reuse cluster is needed.

For sectoring, replace the six-interferer sum with the directional interference sum: $SIR = G_0 R^{-n} / \Sigma_i G_i d_i^{-n}$, where G_0 is serving-sector gain and G_i is the gain of each interfering sector toward the mobile. A simplified two-dominant-interferer approximation is

SIR ≈ (D/R)^n/2, but a real calculation must use measured antenna patterns and all significant interferers.

3.5 Capacity and cost relation

If C_B(48,0.02) is the Erlang-B traffic supported by 48 channels at 2% blocking, the ideal system traffic capacity for an N-cell reuse cluster across the whole area is approximately C_sys = M × C_B(■336/N■,0.02), where M = ceil(A_town/A_cell). For fixed M, increasing N reduces channels per cell and therefore per-cell trunking capacity, while improving SIR. Reducing R increases M approximately in proportion to $1/R^2$ and increases spatial reuse, but adds sites.

A simple lifecycle planning-cost model is C_plan = C_survey + M C_site + C_optimisation + C_backhaul. Define cost per carried Erlang as C_unit = C_plan/C_sys. Fixed reuse has lower site/optimisation cost but may have lower feasible capacity; sectoring adds antenna and optimisation cost; macro–micro adds site/backhaul cost but concentrates investment where traffic exists. This gives the required cost–capacity relationship without pretending that unprovided site prices are known.

For the worked geometry: 19 macro cells × nominal 48 channels/cell is a first-order allocation view. Because cluster channel groups repeat spatially, the same 336-channel block is reused across the grid; total simultaneous traffic capacity depends on number of cells, interference feasibility and Erlang-B loading, not simply 336 calls for the entire township.

TASK 4 – Best, Average and Worst-Case Behaviour (4 Marks)

Best SIR occurs close to the serving base station because desired path loss is small. Worst SIR generally occurs near a cell boundary/corner facing co-channel cells, where desired distance is large and several interferers may be relatively close. Mid-cell positions are intermediate. The six-equidistant-interferer expression is a conservative geometry simplification for an edge user; actual mobile position and antenna direction change each distance.

Illustrative distance calculation for N=7, R=0.5 km, n=3.5: at the cell centre, desired distance may be approximated by 0.1 km; at mid-cell use 0.25 km; at the edge use 0.5 km. If six interferers are each approximated at D=2.291 km, SIR ratio = (D/d_serv)^n/6. This gives approximately 42.9 dB at 0.1 km, 29.0 dB at 0.25 km and 18.5 dB at 0.5 km. These are idealised illustrative positions, not a site-specific prediction.

Case	Mobile position / traffic time	SIR / blocking behaviour	Reason
Best	Near base station; off-peak	High SIR; very low blocking	Strong desired signal; spare channels
Average	Typical mid-cell; average busy hour	Moderate SIR; blocking near target only in loaded cells	Mixed geometry and moderate occupancy
Worst	Cell corner / boundary; event busy hour	Lowest SIR; highest outage and blocking	Weak desired signal, several interferers, channel pool saturation

Blocking is driven mainly by offered load and the finite number of channels, not by mobile position alone. Off-peak demand gives low blocking; busy-hour demand raises occupancy; an event surge can exceed local channel capacity. Shadowing can attenuate the desired signal or an interferer unpredictably. A larger path-loss exponent reduces distant interference more rapidly, often improving reuse SIR, but dense buildings can also create deep local shadowing and coverage holes. Hence n = 2.5 open-campus and n = 4 dense-built-up cases should both be tested.

TASK 5 – Scaling and Sensitivity Analysis (3 Marks)

Key proportionalities: channels per cell ≈ S/N; reuse distance $D \propto R\sqrt{N}$; under the six-interferer approximation SIR ∝ N^(n/2); cell count $M \propto 1/R^2$; and offered traffic A_total ∝ subscriber count. Sectoring reduces the effective interference sum, while microcell splitting increases the number of spatially reusable cells.

Subscribers	Offered traffic (Erlangs)	Average traffic/cell (19 cells)	Planning implication
1,000	25	1.32	Macro-only baseline is ample on average
10,000	250	13.16	Monitor demand hotspots and busy-hour distribution

Subscribers	Offered traffic (Erlangs)	Average traffic/cell (19 cells)	Planning implication
20,000	500	26.32	Sectoring and selective microcells become useful
100,000	2,500	131.58	Densification and dynamic load management required

As n rises from 2.5 to 4, distant co-channel interference falls more sharply, improving theoretical SIR for the same N; however, dense buildings increase shadowing and non-line-of-sight variability. A practical break-even is reached when the busiest cells approach their Erlang-B 2%-blocking load or when the 5th-percentile SIR falls below the target. At that point, sectorise or split only the affected cells. A universal subscriber threshold is not defensible without a traffic map; 20,000 subscribers is a sensible trigger for detailed hotspot review, while the 100,000 case clearly requires a denser architecture under the stated average traffic.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

Let M■ be the initial number of cells and let R■ be the original radius. A split replaces one parent cell by four child cells of radius R/2 (idealised area conservation: $4 \times (R/2)^2 = R^2$). Let f_k be the fraction of cells split in round k, where $0 \le f_k \le 1$. Then $M_k = M_{(k-1)}(1 + 3f_k)$. Uniform splitting has $f_k = 1$, so $M_k = 4M_{(k-1)}$ and $M_k = M■4^k$.

If per-cell offered capacity is C■ and each child retains the same channel plan through spatial reuse, ideal aggregate capacity scales as $C_k = C■4^k$ for uniform splitting, subject to interference and finite channel constraints. For selective splitting, $C_k = C_{(k-1)}(1 + 3f_k)$ in the ideal area-proportional model.

The question gives the power relation P′ = P/2■. Thus for uniform splitting, $P_k = P■/2^{(nk)}$; for a selectively split cell, child transmit power is P_parent/2■. This is a simplified link-budget scaling, not a universal radio rule; actual power is set by coverage, antenna gain and interference constraints. Handoff load rises as cell radius shrinks because users cross more cell boundaries per unit distance.

Work/cost recurrence: if one new site costs C_site and each split operation costs C_split, then $W_k = W_{(k-1)} + 3f_kM_{(k-1)}C_{site} + f_kM_{(k-1)}C_{split}$. Base case: M■ = 37 (geometry estimate), R■ = 0.5 km, P■ as the original macro transmit power, and C■ as the baseline Erlang-B capacity.

TASK 7 – Solve the Recurrence and Interpret the Result (4 Marks)

For uniform splitting, $M_k = 4M_{(k-1)}$, M■ known. Repeated substitution gives M■ = 4M■, M■ = 4²M■, and therefore $M_k = M■4^k$. Similarly, $C_k = C■4^k$ and $P_k = P■2^{(-nk)}$.

If the demand per original cell is D and the original supported capacity is C■, require $C■4^k \ge D$. Dividing by C■ and taking logarithm base 4 gives k ≥ log■(D/C■). Since rounds are integers, k_min = max(0,■log■(D/C■)■). This matches the requested expression.

For selective splitting, if only a fixed fraction f of cells is split each round, $M_k = M■(1+3f)^k$ and $C_k = C■(1+3f)^k$ under the same ideal assumptions. Therefore k_min = max(0,■ln(D/C■)/ln(1+3f)■). If only a particular hotspot is overloaded, apply the recurrence to that hotspot's local demand and capacity rather than multiplying the whole township.

With n = 3.5, uniform splitting reduces the child power per round by $2^{3.5} \approx 11.31$ relative to the parent in this simplified model. Site count grows fourfold per round and handoff frequency increases as radius halves. This shows why selective splitting is usually more economical than uniform splitting.

TASK 8 – Implement and Test the Planning Strategies (4 Marks)

The following compact Python reference simulator provides a reproducible starting point. It calculates valid cluster sizes, reuse distance, the six-interferer SIR approximation, Erlang-B blocking, and a simple traffic/capacity screen for the four subscriber scenarios. It is not a full Monte-Carlo hex-grid ray-tracing simulator; a complete field-grade model must add random mobile positions, shadowing, measured antenna patterns, channel assignments and finite-grid geometry.

```
import math
def erlang_b(A, m):
    B = 1.0
    for k in range(1, m + 1):
        B = (A * B) / (k + A * B)
    return B

def valid_clusters(limit=20):
    return sorted({i*i + i*j + j*j
                   for i in range(limit) for j in range(limit)
                   if 0 < i*i + i*j + j*j <= 20})

def sir_db(N, n, interferers=6):
    sir = (3*N)**(n/2) / interferers
```

```
    return 10 * math.log10(sir)

def channels_per_cell(total_channels, N):
    return total_channels // N

def required_channels(A, target=0.02):
    m = 1
    while erlang_b(A, m) > target:
        m += 1
    return m

S = 336
for subscribers in [1000, 10000, 20000, 100000]:
    A = subscribers * 0.025
    print(subscribers, "users:", A, "Erlangs")
    for N in [3, 4, 7, 12]:
        m = channels_per_cell(S, N)
        print(" N=", N, "channels/cell=", m,
              "SIR n=4=", round(sir_db(N, 4), 2), "dB",
              "blocking at average 19-cell load=",
              round(erlang_b(A/37, m), 4))
```

Test matrix required by the assignment: N = 3, 4, 7, 12; n = 2.5, 3, 4; omnidirectional and 3-sector/6-sector antenna cases; subscribers = 1,000, 10,000, 20,000 and 100,000. Use fixed random seeds for every strategy so that the same mobile locations and shadowing samples are compared. The code above is a deterministic sanity-check core; it should be extended before claiming Monte-Carlo experimental completion.

TASK 9 – Record and Present Experimental Performance (3 Marks)

The required comparison metrics are: SIR mean and 5th percentile, outage probability below the selected SIR threshold, Erlang-B blocking probability, channels per cell, sites required and simulator run-time. Report identical traffic inputs and random seeds for each strategy.

Metric	How to calculate / present	Why it matters
SIR distribution	Mean, 5th percentile and CDF	Shows typical and edge-user radio quality
Outage probability	Fraction of samples with SIR below threshold	Measures coverage/interference failure
Blocking probability	Erlang-B per cell/sector or simulated call loss	Measures access capacity
Channels per cell	floor(336/N), with sector allocation stated	Shows spectrum sharing
Sites required	Area / hex-cell area, plus densification sites	Captures deployment cost
Run-time	Wall-clock time and hardware/software configuration	Shows computational scalability

No simulation was executed as part of this written solution, so no fabricated SIR percentiles, outage rates, blocking measurements or run-times are reported. The analytical screening values above are calculations. After running the code and extending it to Monte-Carlo geometry, insert measured tables and SIR CDF/blocking-versus-load plots here. The chart below shows offered traffic growth calculated directly from the assignment traffic assumption, not simulation output.

TASK 10 – Compare Theory with Simulation (4 Marks)

The analytical six-interferer SIR model predicts the trend that larger N and larger n improve SIR, while channels per cell fall as N increases. A spatial simulator should broadly reproduce those trends. Differences are expected because the analytical model assumes six equal-distance interferers, ideal hexagons and simplified propagation.

Source of difference	Effect on comparison
Equidistant-interferer approximation	Real interferer distances and antenna gains differ; SIR distribution replaces one nominal value
Finite tiers / finite grid	Boundary cells have fewer visible interferers unless wrap-around is used
Mobile-position distribution	Cell-centre users have stronger desired signal than edge/corner users

Source of difference	Effect on comparison
Shadowing and multipath	Randomly raises or lowers desired/interfering powers
Erlang-B assumptions	Poisson arrivals, exponential holding times and infinite-source approximation may differ from measured call patterns
Integer rounding	floor(336/N) and indivisible channels change capacity relative to continuous formulas

A fair validation process is: (1) compare simulator mean/median SIR with the analytical reference for the same geometry; (2) check the N=7, n=4, six-interferer result near 18.7 dB; (3) add realistic mobile locations and antenna patterns; (4) compare blocking from Erlang-B with call-level simulation; and (5) explain every residual gap. Since no run output is available here, the final ranking is a reasoned design expectation, not a claimed experimental finding.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommended architecture: a macro coverage layer with an N = 7 starting reuse plan, three-sector antennas in high-density academic/hostel/residential zones, and selectively deployed microcells in hotspots. Use coordinated dynamic channel assignment/borrowing where adjacent-cell interference checks pass. Keep control channels protected and coordinate neighbour lists, power, tilt and handoff thresholds.

Parameters: R = 0.5 km is the initial planning radius; N = 7 is a baseline rather than a final universal value; n = 2.5–4 must be mapped by environment; design SIR target = 18 dB for conservative planning; busy-hour load = 0.025 Erlang/subscriber; blocking target = 2%. Because the simplified unsectored SIR for N=7,n=3.5 is only about 16.1 dB, sectorisation or more conservative reuse is needed in areas where the 18 dB target is mandatory.

Why this is preferred: fixed reuse is inexpensive but inflexible; sectorisation improves interference control without increasing spectrum; selective microcells add capacity where it is needed rather than everywhere. This balances trunking efficiency, interference, site cost and mobility. Beyond 100,000 subscribers, the macro-only layout is not practical with the stated traffic. A dense small-cell layer, more sectors and careful mobility management are required; the fixed spectrum remains the limiting resource.

TASK 12 – Day-4 Design Challenge: Stadium Event (8 Marks)

A. Impact Analysis (2 Marks)

Event offered traffic = 25,000 attendees × 0.05 Erlang/person = 1,250 Erlangs in 1 km². Event traffic density is therefore 1,250 Erlangs/km². The township's normal busy-hour traffic density at 100,000 subscribers is 2,500/12 = 208.3 Erlangs/km² on average. The event zone is about six times that average density; compared with the 20,000-subscriber township average (500/12 = 41.7 Erlangs/km²), it is about 30 times higher.

The assumptions of spatially uniform and steady demand no longer hold. Demand is concentrated in a stadium, begins and ends around a known event, and lasts four hours. Predictability and short duration favour temporary capacity rather than permanent township-wide cell splitting.

B. Evaluation of Original Design (2 Marks)

The original macro-only N=7 plan has approximately 48 channels per cell. A single 48-channel cell cannot support 1,250 Erlangs at 2% blocking. Erlang-B requires a much larger channel pool; even with a 336-channel pool concentrated entirely in the event zone, it is necessary to calculate blocking and recognise that the entire township would then lose its ordinary channel allocation. Therefore, unchanged reuse is not acceptable. The first failure is local channel congestion/blocking; interference and access-channel congestion can also worsen as users transmit near the stadium.

An event-specific 1 km² zone needs multiple coordinated sectors/small cells. The ideal geometric cell count at R=0.5 km is about 1/0.325 = 3.08 cells per km², but capacity rather than geometric coverage drives the event redesign. The 1,250-Erlang demand is the governing surge figure.

C. Adaptation / Redesign (2 Marks)

Choose MODIFY temporarily. Keep the permanent macro network. For event days, deploy a temporary cell-on-wheels or temporary small-cell cluster around the stadium, split the stadium into directional sectors, and use dynamic channel allocation/borrowing from neighbouring low-load cells under an interference constraint. Add temporary backhaul and power, preconfigure neighbour relations and handoff thresholds, and remove or power down temporary sites after the four-hour event.

With four-way splitting, each child radius is R/2 and the ideal site count in the stadium zone increases fourfold; the local area capacity can also increase approximately fourfold if the channel reuse/interference plan permits. Sectoring reduces the number and strength of dominant

interferers. The cost is temporary site rental, backhaul, engineering and event operations; it is likely cheaper than permanently densifying the entire township for two events per month.

D. Evidence-Based Defence (2 Marks)

The event creates 1,250 Erlangs/km² for four hours, far above the average township density. A permanent redesign would impose recurring cost and additional handoffs for a demand spike that occurs only about twice monthly. Temporary stadium cells plus sectoring and controlled channel borrowing target capacity at the location and time where it is needed. The design must still be validated with a detailed event traffic forecast, per-sector Erlang-B calculation, interference simulation and backhaul test. Retain the permanent macro layer, modify temporarily for event days.

Conclusion

A 336-channel fixed allocation can be reused spatially, but reuse distance, interference and per-cell trunking capacity must be balanced. N=7 provides a defensible baseline; sectoring and selective microcell splitting improve capacity in dense areas. Traffic rises from 25 Erlangs at 1,000 users to 2,500 Erlangs at 100,000 users, so a macro-only design does not scale indefinitely. Repeated cell splitting increases cell count and ideal spatial capacity while reducing transmit power and increasing handoff load. For the predictable stadium surge, temporary sectorised small cells are the most proportionate response. All numerical design values are planning estimates; final deployment requires measured propagation, a detailed traffic map and completed simulator results.

References and Technical Notes

1. Assignment brief: ITA03-MC Assignment 1, Mobile Computing, CO1 (provided source document).

2. W. C. Y. Lee, Mobile Communications Design Fundamentals, 2nd ed., Wiley.

3. T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall.

4. A. G. Reudink / standard cellular engineering treatments of frequency reuse, Erlang-B traffic and co-channel interference.

5. Erlang-B recurrence and hexagonal reuse equations are standard cellular traffic/radio-planning models; numerical deployment values must be validated against local measurements.

Submission integrity note: Tasks 8–10 request implementation and experimental measurements. This document supplies a reproducible starter implementation and a validation protocol, but does not claim that a full Monte-Carlo experiment was executed. Run and extend the code, then replace the clearly identified analytical/illustrative sections with your actual outputs before submitting if your instructor requires execution evidence.

